High Performance Liquid Chromatography Filtering Standard Operating Procedure

Version 1.0

Authors

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Protocol Revision Record

Version	Date Implemented	Description of Revisions
1.0	10 May 2025	Initial release in format standardized for Bio-GO-SHIP

Acronyms, Abbreviations, Terminology

Term	Definition
Inline/underway water	Sampling water that is pumped from beneath the ship to a receiving spigot in an onboard laboratory
CTD rosette water	Sampling water gathered from Niskin bottles on the conductivity, temperature, density (CTD) rosette
DI water	Deionized water from the Milli-Q system
HPLC	High performance liquid chromatography
GF/F	Glass fiber filter
GO-SHIP	Global Ocean Ship-Based Hydrographic Investigations Program
Bio-GO-SHIP	Biological GO-SHIP

Background

Summary

This protocol describes discrete sampling of algal pigments via HPLC while onboard a research vessel, with sampling water from the CTD rosette or underway system.

Method description and rationale

Algal pigments will be analyzed with HPLC on samples collected from both inline flow through systems and CTD bottle rosettes. Sample volume will vary based on particle or algal load and typically range from 1 to 4 L for eutrophic to oligotrophic conditions, respectively. Sample particulates will be collected via low pressure (less than 5 mm Hg) vacuum filtration onto pre-combusted (450C, 4 hours) 25 mm Whatman GF/F filters. Samples will be analyzed at the NASA Goddard Space Flight Center (GSFC) Ocean Ecology Laboratory following established protocols (Van Heukelem and Thomas, 2001).

Duplicate samples will be collected at a rate of one every ten samples. GSFC's sample processing methods include an assessment of the duplicate samples, sample precision by reinjection of samples, and an effective limit of quantitation (lower detection limit) which are all provided with the data report.

Spatial coverage and environment(s) of relevance

This protocol can be used across any marine environment to effectively collect HPLC data from surface waters and at depth.

Ship space requirements

About 4 square feet of counter space is needed for filtration rigs and vacuum pump, and a waste carboy will be secured nearby on the ground.

Personnel Required

One person that is trained in filtration protocol. Typically, two individuals that operate on 12-hour shifts.

Safety

There are no major safety concerns with this protocol.

Time needed to execute the procedure

To complete protocol from gathering sample water, filtering, and storing samples may take about 30 minutes.

Standard Operating Procedure

Materials:

- 25 mm 0.7 um GF/F filters, pre-combusted at 450 C for 4 hours

- 2L dark sample bottles
- 1 ml cryovials

Filtering instructions:

- Wear gloves when gathering sample water and filtering
- Prep filter frit ahead of time by using forceps to place one combusted GF/F on the frit, reattach filter cup, dampen GF/F with DI water, and cover filter cup with a piece of foil
- Rinse sample bottle 3 x with sample water
- Invert sample bottle over the filter frit and GF/F
- Filter sample in relatively dark room or space, e.g. overhead lights off or covered
- Watch water carefully – let the filter just barely run dry as you stop filtering (HPLC method is sensitive to water on filter but want to avoid pulling sample dry and disrupting cells)
- Use forceps to fold GF/F in thirds or quarters, sample side folded inwards. avoid touching the sample area
- Store filter in a labeled cryovile
    - Label as: HPLC, date, station number, volume filtered, initials
    - Include A, B, or C if there are replicates at the same station
- Duplicates are required on 10% of samples. Collect a duplicate every 10th sample, or when it is most convenient to do so
- Empty waste carboy
- Fill out elogs

Basic troubleshooting

Contact Jason Graff via WhatsApp for any issues encountered.